Aaryan S. Maralihalli

Machine Learning / Research Engineer · *(also for new-grad Applied Scientist / research-associate programs)* · Bangalore, India · open to remote · aaryanmaralihalli@gmail.com · +91-9019884683 · github.com/Aaryansm11 · linkedin.com/in/aaryan-maralihalli

Summary

ML researcher with a **peer-reviewed IEEE publication** in interpretable deep learning for physiological signals. Builds **self-supervised / contrastive** and multimodal models with explainability, end-to-end from data to evaluation — and ships them (FastAPI). CS graduate, Jul 2026.

Publication

- **Aaryan S. Maralihalli et al.**, *"An Interpretable Model for Stroke Prediction and Arrhythmia Classification using ECG & PPG,"* **IEEE ICASI 2026**, Kyoto, Japan — **Oral Presentation** (Paper J260481).

Research & Projects

Self-Supervised Cardiac Pattern Discovery (capstone — basis of the publication) github.com/Aaryansm11/ecg-ppg-arrhythmia-prediction

- **Contrastive learning (SimCLR, NT-Xent loss, τ=0.07)** on the **MIMIC-III** Waveform DB — 133,157 segments, 8 channels, 288 matched patients; designed a 1D-CNN **WaveformEncoder** (~238K params, 128-d embeddings); fused with a Random Forest clinical model; **ROC-AUC 0.757**, benchmarked against **CHA2DS2-VASc**; explainability via temporal saliency, channel importance, and **integrated gradients**.

FinFusion

- Built the **core multimodal forecasting model** (LSTM + transformer sentiment fusion), MAPE ↑15–20% vs single-modal baselines.

LANGTRON github.com/Aaryansm11/LANGTRON

- Fine-tuned **Mistral-7B (LoRA)**; FAISS semantic retrieval.

Experience

HyperVerge, Bangalore
Integration Engineer Intern Jan 2026 – Jun 2026

- Applied production AI/ML verification models (face match, liveness, name match) for enterprise fintech clients; drove model-scoring improvements.

Integrated Genetic Solutions (IGS), Bangalore
Project Intern, Medical AI Jun – Aug 2024

- ML workflows for medical document processing (HIPAA-compliant).

Skills

- **Research:** self-supervised / contrastive learning, multimodal modeling, signal processing (ECG/PPG), experimental design, explainable AI, scientific writing
- **Tooling:** PyTorch, HuggingFace, NumPy/Pandas, scikit-learn, FAISS, Python

Education

- **B.Tech, Computer Science** — PES University, Bangalore — 2022 – Jul 2026
- *Certifications:* CITI Program (research ethics: Conflicts of Interest; Data/Specimens Only Research)